

Anna French

Department of Economics, W. P. Carey School of Business, Arizona State University

asamodel@asu.edu | annaafrench.com | (480) 803-8937

U.S. Permanent Resident

EDUCATION

Arizona State University

Fall 2021–expected May 2027

PhD in Economics

New York University

Spring 2025

Visiting PhD Student

Arizona State University

Fall 2021–Fall 2022

M.S. in Economics

New Economic School, Higher School of Economics

Fall 2017–Spring 2021

Joint B.A. in Economics

RESEARCH INTERESTS

Macroeconomics; Environmental Economics; Energy Economics; Spatial Economics

WORKING PAPERS

“The Aggregate Impact of Policy Uncertainty in the Green Energy Transition” — *Job Market Paper*

Abstract: Federal subsidies for wind and solar are the central instrument of U.S. electricity sector decarbonization, yet their presence has been historically uncertain. I study the macroeconomic and environmental costs of renewable policy uncertainty in a dynamic spatial general-equilibrium model with a stochastic subsidy and irreversible capital. I compute the transition with and without subsidy uncertainty, calibrating the uncertainty to the U.S. record of single-party control. I find that relative to a budget-neutral but uncertain subsidy, a permanent subsidy adds 13% more renewable capacity, which displaces gas, reducing cumulative power-sector emissions by 3.2% (2.29 Gt). At the same budget, making the subsidy credible generates an aggregate welfare gain with a present value of \$322 billion. The uncertainty cost depends on the policy reversal being unforeseen; a repeal with a known date causes renewable capacity to be installed inefficiently early. Uncertainty has the greatest effect in the regions that are least suitable to wind and solar generation, like the Midwest, where renewable adoption depends on subsidies.

“Spatial Effects of Diesel Tax: Trade and Externalities” — Submitted

Abstract: Freight transported by heavy-duty trucks generates carbon, local pollution, congestion, accident, and pavement damages along their routes. I develop a multi-commodity open-economy trade model that follows each trade flow through the U.S. highway network, allocating emissions, damages, and tax revenue to every state and county it crosses. Using Freight Analysis Framework shipment data and web-scraped diesel prices, I estimate freight responses to fuel costs. Increasing the federal diesel tax to its 1993 real value avoids more than \$700 million in damages over 2018–2024, with pavement damage generating the largest benefit. Most benefits occur in corridor states that freight passes through, and part of the welfare cost is borne by foreign buyers of U.S. exports. A unilateral state diesel tax creates large spillovers: when New Mexico raises its tax to California’s level, freight shifts to alternative corridors, increasing externalities in neighboring states.

“Permits as Real Options: Evidence and Implications for Regulatory Leakage”

(with Nicholas Vreugdenhil) — Submitted

Abstract: We identify a new channel for regulatory leakage: the “permitting channel.” In many settings, permits lock in the regulatory regime at approval, allowing firms to acquire the option to invest under current rules. Unlike the conventional Green Paradox response, this channel operates without accelerating physical investment and pollution. We develop a framework in which firms can use real options to avoid anticipated regulation. We estimate the framework using Colorado’s oil and gas reforms, where permits surged before regulation but drilling did not. Policy-relevant designs that remove the permitting channel substantially reduce the health risks from pollution despite amplifying anticipatory investment.

WORK IN PROGRESS

“The Forest Is Worth More Standing Than Cut” (with Wyatt Brooks)

AWARDS AND FELLOWSHIPS

AEA Summer Fellowship, Washington Center for Equitable Growth	<i>Summer 2026</i>
Best Progress Towards Dissertation Award, Arizona State University	<i>Fall 2025</i>
Most Deserving Graduate Student in the Program, Arizona State University	<i>Spring 2024</i>
Best Oral Presentation (Comprehensive Exam), Arizona State University	<i>Spring 2024</i>
Travel Fellowship, Environmental and Energy Economics Summer School, UC Berkeley	<i>Summer 2023</i>
Best Macroeconomics Qualifying Exam, Arizona State University	<i>Summer 2022</i>
Outstanding Female Fellowship, Arizona State University	<i>Fall 2021</i>
National Finalist (Top 12), All-Russian Olympiad of School Students in Economics	<i>Spring 2017</i>

TEACHING EXPERIENCE

Instructor , Arizona State University	<i>Fall 2025</i>
Intermediate Microeconomic Theory (ECN 312)	
Teaching Assistant , Arizona State University	<i>Fall 2022–Spring 2026</i>
Topics in Environmental Economics (ECN 791, ECN 445), Microeconomic Principles (ECN 212), Economics of Gender & Family (ECN 349)	

PRESENTATIONS

- 2026** Midwest Macroeconomics Meeting (Texas Tech University)*; Arizona Economics Job Market Candidates Mini-Conference (University of Arizona)*; PhD Reunion Conference (Arizona State University)*; Modeling Climate-Dynamics Workshop for Young Scholars (University of Chicago)
- 2025** Midwest Macroeconomics Meeting (Federal Reserve Bank of Kansas City); Environmental & Resource Economics Workshop (University of Colorado); AZ Workshop on Environment, Natural Resource, and Energy Economics (University of Arizona); Center for Environmental Economics and Sustainability Policy Seminar (Arizona State University)
- 2024** Midwest Macroeconomics Meeting (Virginia Commonwealth University)
- 2023** AZ Workshop on Environment, Natural Resource, and Energy Economics (Arizona State University)
- Internal:* ASU Macroeconomics Workshop (2024–2026); ASU General Economics Workshop (2023)

* upcoming

RESEARCH EXPERIENCE

Research Assistant , Arizona State University Wyatt Brooks (2023–present); Nicholas Vreugdenhil (Summer 2024, 2025); Nicolai Kuminoff (Summer 2024); Kelly Bishop (Summer 2023)	<i>Summer 2023–present</i>
Research Assistant , Higher School of Economics Madina Karamysheva	<i>Fall 2020</i>
Research Assistant , Tufts University Christopher R. Miller	<i>Spring 2019</i>
Research Assistant , New Economic School Andrei Markevich	<i>Fall 2018–Fall 2019</i>

OTHER INFORMATION

Programming Skills: Julia, Python, R, Stata, MATLAB, LaTeX
Languages: English (fluent), Russian (native), French (intermediate)

REFERENCES

Wyatt Brooks (Advisor) Associate Professor Arizona State University wyattjbrooks@gmail.com	Nicolai Kuminoff Associate Professor Arizona State University kuminoff@asu.edu
Nicholas Vreugdenhil Assistant Professor Arizona State University nvreugde@asu.edu	